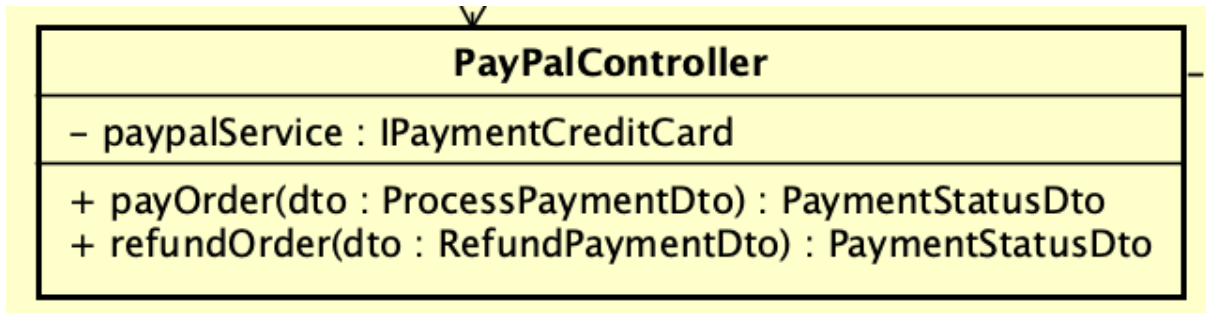


CLASS SPECIFICATION FOR PAYPAL BY CREDIT CARD

Design for class "PayPalController"



Description: PayPalController is the entry class of the PayPal Integration subsystem. It receives payment or refund requests from the outer layer and delegates the actual processing to a service that realizes credit-card payment behavior. This class keeps the subsystem boundary simple and exposes only the business operations needed by the caller.

Table 1. Example of attribute design

#	Name	Data type	Default value	Description
1	paypalService	IPaymentCreditCard	Injected dependency	Service object used to process credit-card payment and refund requests through the PayPal subsystem.

Table 2. Example of operation design

#	Name	Return type	Description (purpose)
1	payOrder(dto : ProcessPaymentDto)	PaymentStatusDto	Accepts a payment request, delegates the processing to the bound payment service, and returns the payment status result for the caller.
2	refundOrder(dto : RefundPaymentDto)	PaymentStatusDto	Accepts a refund request and delegates it to the payment service to perform refund processing through PayPal.

Parameter:

- dto: no default value, contains orderId, amount, transactionContent, cardNumber, expDate, owner and other information required to create and execute a PayPal order.
- dto: no default value, contains transactionId, amount, orderId and refund reason needed to request a PayPal refund.

Exception:

- PaymentException if the delegated service reports a payment-related error, such as invalid request data, failed order creation, capture failure, or refund failure.

Method

- This class does no gateway-specific processing. It forwards validated request DTOs to paypalService and returns the service result unchanged to preserve separation of concerns.

Other diagrams

- Related diagram: Sequence diagram "Pay order by credit card".
- State diagram: not required for this controller.

Design for class "PaypalService"

PaypalService
- paypalApiClient : PaypalApiClient + processPayment(dto : ProcessPaymentDto) : PaymentStatusDto + refund(dto : RefundPaymentDto) : PaymentStatusDto + checkPaymentStatus(dto : CheckPaymentStatusDto) : PaymentStatusDto - buildPaypalOrderReq(dto : ProcessPaymentDto) : PaypalOrderRequestDto - mapToPaymentStatus(paypalRes : PaypalCaptureResponseDto) : PaymentStatusDto

Description: PaypalService contains the core business logic of the PayPal Integration subsystem. It converts generic payment requests into PayPal-specific request DTOs, obtains gateway responses through PayPalApiClient, maps those responses back to the system payment status representation, and supports payment, refund, and payment-status checking behaviors.

Table 1. Example of attribute design

#	Name	Data type	Default value	Description
1	paypalApiClient	PaypalApiClient	Injected dependency	Low-level client used to communicate with the PayPal REST APIs for token retrieval, order creation, payment capture, and refund execution.

Table 2. Example of operation design

#	Name	Return type	Description (purpose)
1	processPayment(dto : ProcessPaymentDto)	PaymentStatusDto	Builds a PayPal order request from generic payment data, requests PayPal order creation/capture through the API client, and

			returns the mapped payment status.
2	refund(dto : RefundPaymentDto)	PaymentStatusDto	Uses PayPal capture/refund information to request a refund and converts the gateway result into system payment status.
3	checkPaymentStatus(dto : CheckPaymentStatusDto)	PaymentStatusDto	Checks the current payment state of a previously created transaction and returns a status object understood by the rest of the system.
4	buildPaypalOrderReq(dto : ProcessPaymentDto)	PaypalOrderRequestD to	Creates a PayPal-specific order request object from the generic payment data required by the subsystem.
5	mapToPaymentStatus(paypalRes : PaypalCaptureResponseDto)	PaymentStatusDto	Transforms a PayPal capture response into the generic payment status object used by AIMS.

Parameter:

- dto: no default value, includes the order identifier, transaction content, payment amount, return/cancel context, and card-owner-related information required to prepare a PayPal payment request.
- dto: no default value, includes refund amount, orderId, transactionId and refund reason.
- dto: no default value, includes at least orderId and transactionId for transaction lookup.
- dto: no default value, source object used to populate intent, purchase units, return URL, cancel URL and description of the PayPal order request.
- paypalRes: no default value, PayPal capture response to be translated into system-level success, message and status information.

Exception:

- PaymentException if access token retrieval fails, PayPal order creation fails, payment capture fails, refund is rejected, or the gateway returns an unknown payment error.

Method

- processPayment() should first obtain an access token, then build the PayPal order request, call createOrder(), extract the order identifier or approval context if needed, call capturePayment(), and finally convert the gateway response into PaymentStatusDto.
- refund() should use stored transaction/capture information from the request context, call refundPayment(), and map the refund outcome into a generic payment status.
- Private helper methods are used to keep gateway-specific payload construction and response mapping inside the subsystem.

Other diagrams

- Related diagram: Sequence diagram "Pay order by credit card".
- No dedicated state diagram is shown in the submitted design.

Design for class "PayPalApiClient"

PayPalApiClient	
- httpService : HttpService - configService : ConfigService - baseUrl : string - clientId : string - clientSecret : string	
+ getAccessToken() : string + createOrder(data : PaypalOrderRequestDto, token : string) : PaypalOrderResponseDto + capturePayment(captureReq : PaypalCaptureRequestDto, token : string) : PaypalCaptureResponseDto + refundPayment(captureId : string, data : any, token : string) : any	

Description: PayPalApiClient is the technical gateway adapter of the subsystem. It encapsulates HTTP communication with PayPal, manages credentials and base URL configuration, obtains OAuth access tokens, sends order creation requests, captures approved payments, and triggers refunds when requested by upper layers.

Table 1. Example of attribute design

#	Name	Data type	Default value	Description
1	httpService	HttpService	Injected dependency	HTTP abstraction used to send requests to the PayPal REST endpoints.
2	configService	ConfigService	Injected dependency	Configuration provider used to load API base URL and client credentials.
3	baseUrl	string	Configured at startup	Base endpoint of the PayPal sandbox or target REST API.
4	clientId	string	Configured at startup	Client identifier used to obtain OAuth access tokens from PayPal.

5	clientSecret	string	Configured at startup	Secret credential paired with clientId for token retrieval.
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Table 2. Example of operation design

#	Name	Return type	Description (purpose)
1	getAccessToken()	string	Requests an OAuth access token from PayPal and returns the token string used in subsequent API calls.
2	createOrder(data : PaypalOrderRequestDto, token : string)	PaypalOrderResponseDto	Sends a PayPal order creation request and returns the parsed gateway response.
3	capturePayment(captureReq : PaypalCaptureRequestDto, token : string)	PaypalCaptureResponseDto	Calls the PayPal capture endpoint for an approved order and returns the capture result.
4	refundPayment(captureId : string, data : any, token : string)	any	Sends a refund request for a captured payment and returns the gateway response object.

Parameter:

- none.
- data: no default value, PayPal order request payload.
- token: no default value, bearer token returned by getAccessToken().
- captureReq: no default value, contains the PayPal order identifier to be captured.
- token: no default value, bearer token used for authorization.
- captureId: no default value, PayPal capture identifier of the payment to be refunded.
- data: no default value, refund request payload sent to PayPal.
- token: no default value, bearer token used for authorization.

Exception:

- PaymentException if HTTP communication fails, configuration is missing, authentication fails, or the PayPal API returns an invalid or unexpected response.

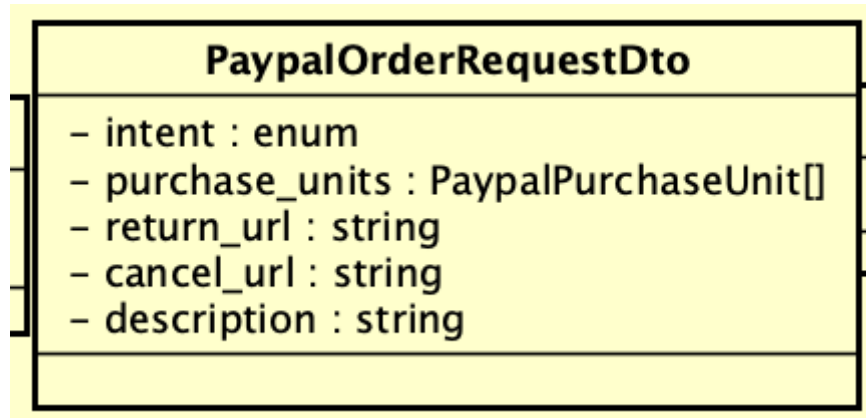
Method

- Credentials and endpoint configuration should be read through configService. Every create/capture/refund call must include the access token returned by getAccessToken(). This class should remain infrastructure-focused and should not contain business status mapping logic.

Other diagrams

- Related diagram: internal subsystem collaboration for the PayPal payment flow can be derived from the submitted subsystem class diagram and the sequence diagram.

Design for class "PaypalOrderRequestDto"



Description: PaypalOrderRequestDto is the request data object used to create a PayPal order. It stores the gateway-specific payload prepared by PaypalService before the request is sent by PayPalApiClient.

Table 1. Example of attribute design

#	Name	Data type	Default value	Description
1	intent	enum	No default value	Requested PayPal order intent, typically indicating the business action of the order.
2	purchase_units	PaypalPurchaseUnit[]	Empty or constructed from request	List of purchase units that carry amount and order-related information required by PayPal.
3	return_url	string	No default value	URL to which the flow is redirected after approval.
4	cancel_url	string	No default value	URL to which the flow is redirected when payment is cancelled.
5	description	string	Empty string or generated text	Human-readable description of the payment or order

				shown in the PayPal context.
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Table 2. Example of operation design

#	Name	Return type	Description (purpose)
1	N/A	N/A	This DTO is a passive data carrier and does not define business operations in the submitted class diagram.

Parameter:

- No operation parameter is specified because this class acts as a passive DTO in the submitted design.

Exception:

- No class-specific exception. Validation is performed by the creator or consumer of this DTO.

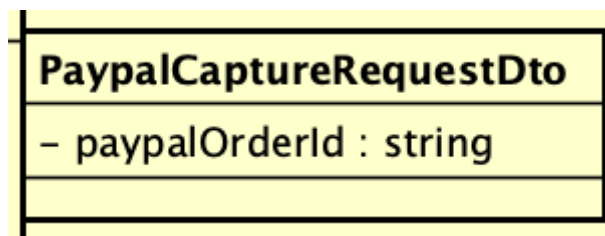
Method

- Instances of this DTO are created by PaypalService from a generic payment request before being sent to PayPalApiClient.createOrder().

Other diagrams

- Related diagram: subsystem class diagram only.

Design for class "PaypalCaptureRequestDto"



Description: PaypalCaptureRequestDto is a compact request object used to capture a previously created PayPal order. It isolates the minimal data needed by the capture API call from the rest of the subsystem.

Table 1. Example of attribute design

#	Name	Data type	Default value	Description
1	paypalOrderId	string	No default value	Identifier of the PayPal order that has been created or approved and is ready to be captured.

Table 2. Example of operation design

#	Name	Return type	Description (purpose)
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1	N/A	N/A	This DTO is a passive data carrier and does not define business operations in the submitted class diagram.
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Parameter:

- No operation parameter is specified because this class acts as a passive DTO in the submitted design.

Exception:

- No class-specific exception. Invalid order identifiers are handled when the request is consumed by PayPalApiClient.

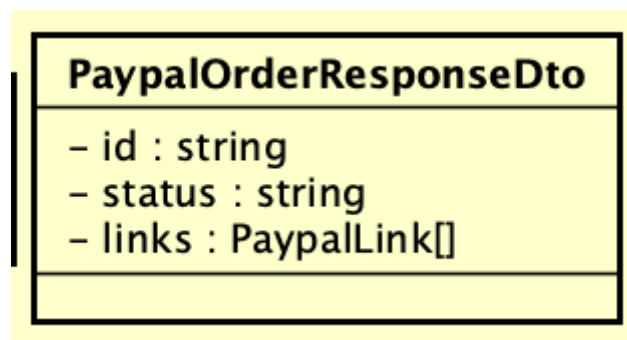
Method

- This DTO is usually created after order creation or approval, then passed to PayPalApiClient.capturePayment().

Other diagrams

- Related diagram: sequence diagram "Pay order by credit card" for the capture step.

Design for class "PaypalOrderResponseDto"



Description: PaypalOrderResponseDto stores the response returned by PayPal after order creation. It gives the subsystem access to the PayPal order identifier, current order status, and navigation or action links supplied by the gateway.

Table 1. Example of attribute design

#	Name	Data type	Default value	Description
1	id	string	Assigned by PayPal	Unique PayPal order identifier returned after successful order creation.
2	status	string	Assigned by PayPal	Current state of the PayPal order returned by the gateway.
3	links	PaypalLink[]	Empty or assigned by PayPal	Collection of PayPal links associated with the order, such as approval or related follow-up actions.

Table 2. Example of operation design

#	Name	Return type	Description (purpose)
1	N/A	N/A	This DTO is a passive data carrier and does not define business operations in the submitted class diagram.

Parameter:

- No operation parameter is specified because this class acts as a passive DTO in the submitted design.

Exception:

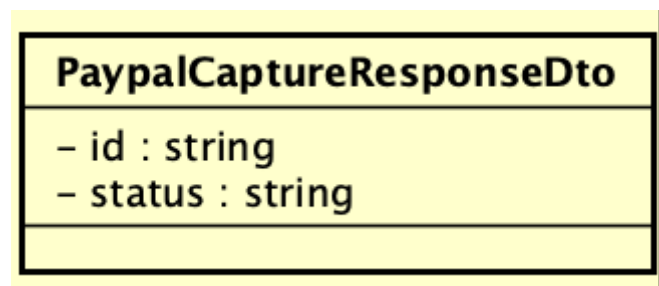
- No class-specific exception. Invalid or incomplete response content is handled by the API client and service layers.

Method

- The object is returned by PayPalApiClient.createOrder() and then interpreted by PayPalService to continue the payment flow.

Other diagrams

- Related diagram: sequence diagram "Pay order by credit card" for the order creation stage.

Design for class "PaypalCaptureResponseDto"

Description: PaypalCaptureResponseDto stores the essential result of a PayPal capture request. It is later mapped into the generic payment status object used by the rest of the system.

Table 1. Example of attribute design

#	Name	Data type	Default value	Description
1	id	string	Assigned by PayPal	Identifier of the captured PayPal payment or resulting capture object.
2	status	string	Assigned by PayPal	Capture status returned by PayPal and used to decide whether the payment is successful or failed.

Table 2. Example of operation design

#	Name	Return type	Description (purpose)
1	N/A	N/A	This DTO is a passive data carrier and does not define business operations in the submitted class diagram.

Parameter:

- No operation parameter is specified because this class acts as a passive DTO in the submitted design.

Exception:

- No class-specific exception. Error handling is performed by the API client and service layers when the response is received.

Method

- Returned by PayPalApiClient.capturePayment() and consumed by PayPalService.mapToPaymentStatus().

Other diagrams

- Related diagram: sequence diagram "Pay order by credit card" for the capture/result stage.